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ORIGINAL ARTICLE

Adaptation strategies to combat climate change effect on rice and mustard in Eastern India

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Abstract The negative impact of climate change on crop production is alarming as the demand for food is expected to increase in coming years, at a rate of about 2 percent a year. Wet season rice (*Oryza sativa*) followed by mustard (*Brassica juncea*) is one of the prominent cropping sequences in Eastern India. Descreases in their productivity due to climate change will not only hamper the regional food security but also affect the global economy. Considering the fact, the present study aims to assess the impact of climate change on productivity of wet-season rice and mustard and to evaluate the effectiveness of agronomic adjustment as adaptation options. Crop growth simulation model (CGSM) is a very effective tool to predict the growth and yield of a crop. One CGSM, namely InfoCrop (Generic Crop Model), was calibrated and validated for the said crops for West Bengal State, Eastern India. After validation, the model was used to predict the yield under elevated thermal condition (1 and 3 °C rise over normal temperature). Moreover, the future weather situation as predicted by PRECIS (Providing Regional Climates for Impacts Studies) model was used as weather input of the CGSM and the yield was predicted for ten selected locations of West Bengal for the year 2025 and 2050. It was observed that the average yield reduction of the wet-season rice would be in the tune of about 20.0 % for 2025 and 27.8 % for 2050. The mustard yield of West Bengal may be reduced by 20.0 to 33.9 % for the year 2025 and up to 40 % for 2050. It was concluded that the negative impact of climate change on mustard grown in winter season will be more pronounced compared to wet-season rice. Adjustment of sowing time will be the simplest and effective adaptation option for both rice and mustard. Increased rate of nutrient application can sustain the rice yield under future climate. The older seedling at the time of transplanting of wet-season rice and increased seed rate of mustard were proved less effective.

Keywords Agronomic adaptation · Climate change · Crop growth model · InfoCrop · Mustard · Rice

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1 Introduction

Southeast Asia, where agriculture is a major source of livelihood, is expected to be seriously affected by adverse effect of climate change (IPCC 2007a). Since most of its economy relies on agriculture and natural resources as primary income, climate change has been and will continue to be critical factor affecting its productivity. Rice (*Oryza sativa*) is the most important staple food in the region (IRRI 2006) and any change of its productivity has an immense global impact. In 2010, rice were harvested from approximately 154 million ha worldwide, of which 137 million ha (88 % of the global rice harvested) were in Asia and 48 million ha (31 % of the global rice harvested) were in Southeast Asia alone (FAOSTAT 2012). The rice-based cropping sequence is mostly followed in Southeast Asia. In India, wet-season rice followed by mustard is one of the predominant cropping sequences. Rice covers about 43.8 million hectares land with total production of 99.50 million tons (CRRI 2011). The rapeseed-mustard (*Brassica spp.*) is the second most important oilseed in India with an average productivity of 771.0 kg ha $^{-1}$ and India ranks second after China (Chattopadhyay et al. 2005). Both the crops play important role in food security and national economy.

In recent past, various organisations and scientific community tried to estimate the degree of climate change and its concomitant impacts (Cumhur and Malcolm 2008; Szwed et al. 2010). There are definite indications of climate change and the increment of temperature is the most prominent one (IPCC 2007b; Jung et al. 2002; Balling Jr and Cerveny 2003; Fauchereau et al. 2003). Increase in atmospheric temperature scales down net photosynthetic rate and stomatal conductance which reduce both yield and crop water productivity (Zhang et al. 2011; Guoju et al. 2013). Although it has been estimated that the yield of cereal crops already declined between 2.5 and 3.8 % globally due to climate change in the last three decades, the regional impact is still lacking (Lobell et al. 2011). In Eastern India, the variability of rainfall and monsoon shifting are additional problems apart from temperature rise (Banerjee and Khan 2008). So, the probable change in climatic scenario and weather situation will have a complex impact on crop production, which should be assessed to address the future food security. Limited work has been done on this problem for West Bengal situation (Banerjee 2008).

Crop growth simulation models are very useful and effective tool for assessing the future crop production scenario (Aggarwal and Kalra 1994; Hoogenboom 2000). There are various CGSMs among which some popular names are Decision Support System for Agro-technology Transfer (DSSAT), IRRI rice crop model (ORYZA), Generic crop model developed at Indian Agricultural Research Institute (InfoCrop), etc. In those models, weather data of the projected climate change scenario can be used as weather-input for the impact study (Mall et al. 2004). Climate change scenarios are generally derived from projections of climatic parameters with the help of Global Climate Models (GCMs). The Regional Climate Model (RCM), like Providing Regional Climates for Impacts Studies (PRECIS) can be used along with GCM for adding the details to global projections (Jones et al. 2004).

In the present research work, InfoCrop model is used to predict the impact of climate change on rice and mustard and to evaluate possible adaptation options. Initially the model is calibrated and validated for the said crops in the study region. The paper has two broad objectives. 1) to assess the impact of elevated temperature and future climatic scenario on rice and mustard productivity. 2) to evaluate the effectiveness of agronomic adjustment as adaptation options.

2 Materials and methods

2.1 Study area

For calibration and validation of crop growth model, field experiments were carried out in the Research Farm of Bidhan Chandra Krishi Viswavidyalaya (BCKV, State Agricultural University), Kalyani (22.57°N, 88.20°E and 7.8 m above mean sea level), West Bengal, Eastern India, which is situated under New Alluvial Agro-climatic Zone. For the impact study, ten stations covering five agro-climatic zones of West Bengal were selected (Table 1). Only the Hill Zone of West Bengal was not included in the study due to unavailability of input data required for CGSM. The characteristic features of all the six zones of West Bengal are also listed in Table 1.

2.2 Crop growth model description

In the present study, InfoCrop model was used to simulate the effect of weather, soil and crop management factors on the growth and yield attributes of wet season rice and mustard. As the model is specially designed for Indian sub-continent (Aggarwal et al. 2006), the same was used in the present study instead of DSSAT, ORYZA, WOrld FOod STudies (WOFOST) or Agricultural Production Systems Simulator (APSIM). The model is written in FORTRAN Simulation Translator (FTS) programming language. The language is recommended by International Consortium for Agricultural Systems Applications (ICASA) as standard language for systems simulation (Jones et al. 2001). For simulation of InfoCrop model, different experimental data on phenological, physiological and morphological development are needed. The required minimum dataset for the CGSM are enumerated in Table 2. To perform and validate any virtual experiment on InfoCrop, we need the said inputs along with actual experimental data and various crop management data, like seed rate, fertilizer dose, irrigation level, etc.

Table 1 The selected weather stations under different agro-climatic zones of West Bengal, India

Agro-climatic regions of West Bengal	Characteristic features	Weather stations from where data for CGSM was collected
1) Hill zone	Terraced, brown forest, shallow, highly acidic (pH 4 to 6), moderately fertile soil. Temperature range: 8.9 to 14.9 °C Annual rainfall: 3,550 mm	–
2) Terai zone	Plain (Flood prone), Sandy to Sandy loam soil. Temperature range: 12.8 to 32.3 °C Annual rainfall: 2,000 to 3,500 mm	Coochbehar, Jalpaiguri
3) Old alluvial zone	Mostly flat, Loam, deep, mostly neutral soil. Temperature range: 15.1 to 35.3 °C. Annual rainfall: 1,600–1,800 mm	Maldah, Balurghat
4) New alluvial zone	Flat to rolling, Light to heavy, acidic to neutral (pH 5.7) soil. Temperature range: 15.6 to 35 °C. Annual rainfall: 1,200–1,700 mm	Kalyani[a], Krishnanagar
5) Red and laterite zone	Undulating, coarse textured, susceptible to erosion, acidic soil. Temperature range: 14.8 to 37 °C. Annual rainfall: 1,100–1,300 mm	Bankura, Puruliya
6) Coastal and saline zone	Alluvial, fine textured, saline soil. Temperature range: 16 to 34 °C. Annual rainfall: 1,500–1,700 mm	Diamond Harbour, Canning

[a] This station is selected for field experimentation

Table 2 Minimum requirement of the crop and soil parameters for InfoCrop model

Weather	Phenological	Soil
1. Maximum temperature (°C)	1. Date of sowing	1.Soil texture
2. Minimum temperature (°C)	2. Date of 1st emergence	2. Soil moisture content at wilting point
3. Rainfall (mm)	3. Date of transplanting (rice)	3. Soil moisture content at Field capacity
4. Solar radiation (Wm^{-2})	4. Date of 50 % anthesis	
	5. Date of physiological maturity	
	6. LAI at different DAS/DAT (rice)	
	7. Dry matter weight at different DAS/DAT (rice)	
	8. Grain yield	
	9. Biomass yield	

2.3 Database generated for InfoCrop calibration

The actual crop data was generated through field experiment carried out at Kalyani Station. The popular wet-season rice cultivar of Eastern India, namely, *Swarna* was grown with different dates of sowing for consecutive 4 years (2007–2010). Thus twelve separate treatment combination were generated. Data on plant parameters (crop height, leaf area index and biomass), phenological observation (period from sowing to tillering, panicle initiation, anthesis and maturity) and yield were recorded from the experiment field. Six data sets from first 2 years were considered for model calibration and the remaining data sets were used for model validation. The required weather data were collected from Kalyani Meteorological observatory, situated only 50 meter away from the crop field. Soil data were collected from Annual Progress Report of Forecasting Agricultural Output using Space, Agrometeorology and Land base observation (FASAL) Project (FASAL 2013). Crop management practices, like date of sowing, amount and date of irrigation, fertilizer, seed rate, etc., were recorded and considered as input for running the model. With all the input parameters, the predicted yield were simulated and compared with actual yield. The calibration was done by simple iteration method.

2.4 Validation of the model

The performance of the model were evaluated through working out of some statistical parameters, namely, Coefficient of determination (R^2), Root Mean Square Error (RMSE), Standard Error (SE) and others (Fox 1981). The R^2 value measures the linearity between actual and simulated values, whereas the RMSE describes the mean absolute deviation between simulated and observed. Accuracy of forecast is some measures to identify how close the prediction is with the actual value. The accuracy of simulation is characterized by lower RMSE and lower SE. Reliability is another aspect of forecast accuracy. Literally it means the extent to which the prediction can be trusted. One measure of reliability is average bias. The following statistical tools (Gordon and Shykewich 2000) were used in the present study:

(a) Bias is simply the average of the predicted value minus the average observed value.

$$\text{Bias} = \frac{1}{N}\sum_{i=1}^{N} (\text{fi}-\text{Oi}) \quad (1)$$

Here, fi is the predicted yield, Oi is the observed yield and N is the number of observation.

(b) Mean absolute error (MAE) is simply the average of the absolute difference between predicted and observed value.

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}(|fi-Oi|) \tag{2}$$

(c) Standard error (SE) is calculated with the help of comparing actual value (Oi) and predicted value (fi). The equation of standard error of the predicted value is as follows:

$$\mathrm{SE} = \sqrt{\frac{1}{(N-2)}\sum\left(Oi-Oi'\right)^2 - \frac{\left[\sum\left(fi-fi'\right)\left(fi-fi'\right)\right]^2}{\sum\left(Oi-Oi'\right)^2}} \tag{3}$$

fi' & Oi' are the average of $f_{i\text{-}N}$ and $O_{i\text{-}N}$ respectively.

(d) Mean square error (MSE) measures the average of the squares of the "errors." The error is the amount by which the value implied by the estimator differs from the quantity to be estimated.

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(fi-Oi)^2 \tag{4}$$

(e) Root mean square error (RMSE) is simply the root of the MSE value. It is usually best to report the root mean squared error (RMSE) rather than mean squared error (MSE), because the RMSE is measured in the same units as the data, rather than in squared units.

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(fi-Oi)^2} \tag{5}$$

2.5 Assessing impact of climate change and adaptation method

The IPCC has projected temperature increase of 0.5 to 1.2 °C by 2020, 0.88 to 3.16 °C by 2050 for the Indian region depending on the future scenario of development (IPCC 2007a). For Eastern India, the mean temperature increase may be in the tune of 1 °C for 2020 and 3 °C in 2050 for A2 scenario (Boomiraj et al. 2010). Considering these projections, the impact of 1 and 3 °C temperature rise (above normal temperature) on production of wet-season rice and mustard has been assessed for only one station (Kalyani) representing Gangetic West Bengal. The average weather data of Kalyani Observatory for the period of 1980 to 2010 was considered as the normal weather data. Then 1 and 3 °C were added with both maximum and minimum normal temperature to obtain elevated temperature regime. This regime was used to observe the effect of increased temperature on crop production.

To observe the impact of projected climatic scenario, output of regional climate model was considered. The PRECIS output generated from Hadley Centre Coupled Model (HadCM3Q) with 25 km resolution for the year 2025 and 2050 were used as the input weather files of InfoCrop model. The weather parameters of PRECIS output used in InfoCrop model were

maximum and minimum temperature (°C), precipitation (mm) and solar radiation (Wm $^{-2}$). Projected weather data of different stations were taken from nearest grid points. The regional forecasting model indicated a significant linear increasing trend for both maximum temperature (0.2 to 0.6 °C per decade) and minimum temperature (0.2 to 0.5 °C per decade) up to 2050, considering 1981 to 2010 data as baseline. An increasing trend of rainfall was also predicted, whereas no increasing or decreasing trend of solar radiation was projected. The generation of climate change scenario was done earlier by an Asia-Pacific Network Project (Luck et al. 2012).

The percent yield reduction for the years 2025 and 2050 were worked out considering the present average potential yield for wet-season rice as 4,500 kg ha $^{-1}$ and the same for mustard as 1,700 kg ha $^{-1}$. The effectiveness of simple agronomic adjustments was evaluated through CGSM under elevated thermal regime. The following agronomic adjustments were considered:

i) Change in age of seedling of wet-season rice during transplanting time,
ii) Application of higher nutrient dose in wet-season rice,
iii) Change in seed rate for mustard, and
iv) Sowing time adjustment for both the crops.

3 Results

3.1 Calibration and validation of model

To validate the InfoCrop model, the predicted crop yield was compared with actual yield. The InfoCrop Mustard was calibrated and validated earlier by Sarmah et al. (2009) for Eastern India and hence not presented here. The simulated yield generated from the calibrated InfoCrop Rice model was compared with actual yield (Table 3). The average predicted yield (4,329.3 kg ha $^{-1}$) was slightly lower than average actual yield (4,353.3 kg ha $^{-1}$). Hence, InfoCrop Rice model under-predicts the yield marginally. The correlation coefficient value (0.94) is significant at 1 % level, the RMSE value is reasonably low (82.0). The SE of predicted grain yield is also considerably low (95.0).

3.2 Projected yield considering 1 and 3 °C temperature rise

The simulated yield of wet-season rice and mustard were observed under normal temperature condition as well as elevated thermal regime (1 and 3 °C rise in temperature over normal) for West Bengal (Table 4). It was considered that rice was sown on 1st June (transplanted 3 weeks after sowing date) and mustard was sown on 22nd October. Both the dates are normal sowing time in the study region. Temperature increment by 1 °C caused yield reduction of wet-season rice by 382.4 kg ha $^{-1}$ (8.7 %). The yield reduction would be more than 900 kg ha $^{-1}$ (21.1 %) for 3 °C rise in temperature. Krishnan et al. (2007) observed −18.0 and −28.2 % changes in potential yield of rice in Eastern India for 2 and 3 °C rise of temperature respectively. Like the present paper, the authors used InfoCrop model (also used ORYZA model, which yielded similar result) and considered that the base level of CO_2 would remain constant (380 ppm). Aggarwal et al. (2006) also used InfoCrop to simulate the grain yield of rice under changing temperature condition and observed about 16.0 % yield reduction for 2 °C temperature rise. In case of mustard, the yield reduction would be 436.3 kg ha $^{-1}$ (25.5 %) for 1 °C rise

Table 3 Measured and simulated yield of wet-season rice for different years

Treatments (Days of sowing)	Forecast (F)	Observed (O)	F-O	Abs (F-O)	(F-O)^2	Within±100 kg	R^2	SE	RMSE
26.05.2009	4,480	4,375	105	105	11,025	0			
09.06.2009	4,710	4,660	50	50	2,500	1			
22.06.2009	4,084	4,105	−21	21	441	1			
26.05.2010	4,637	4,750	−113	113	12,769	0			
09.06.2010	4,212	4,290	−78	78	6,084	1			
22.06.2010	3,853	3,940	−87	87	7,569	1			
	4,329.3 (Average)	4,353.3 (Average)	−24.0 (Bias)	75.7 (ME)	6,731.3 (MSE)	67 % (Accuracy)	0.94	95	82

Table 4 Effect of elevated temperature on yield and maturity of wet-season rice and mustard

Crops	Yield (kg ha^{-1})			Maturity period (days)		
	NT[a]	NT+1 °C	NT+3 °C	NT	NT+1 °C	NT+3 °C
Rice	4,380.0	3,997.6	3,455.2	109	107	104
% yield reduction		8.7	21.1			
Mustard	1,714.0	1,277.7	1,152.3	93	88	85
% yield reduction		25.5	32.8			

[a] NT = Normal temperature

and 562 Kg ha $^{-1}$ (32.8 %) for 3 °C rise in normal temperature. For the similar degree of temperature enhancement, the maturity period will be shortened by 2–5 days in rice and 5–8 days in mustard.

3.3 Probable yield reduction in 2025 and 2050

The average yield reduction of the wet-season rice was observed to be around 20.0 % for 2025. For 2050, the simulation resulted higher yield reduction (27.8 %) compared to 2025 (Fig. 1). Mishra et al. (2013) observed that projected climate change (up to 2040) will have a significant effect on crop production in Eastern India and rice yield may decline up to 22.6 % for lower Gangetic Basin. Similarly, the InfoCrop output showed an average yield reduction of 26.7 and 34.3 % for 2025 and 2050 respectively in case of mustard (Fig. 2). For 2025, the yield reduction of mustard would vary from 20.0 to 33.9 % and the same would be 25.0 to 45.0 % for 2050.

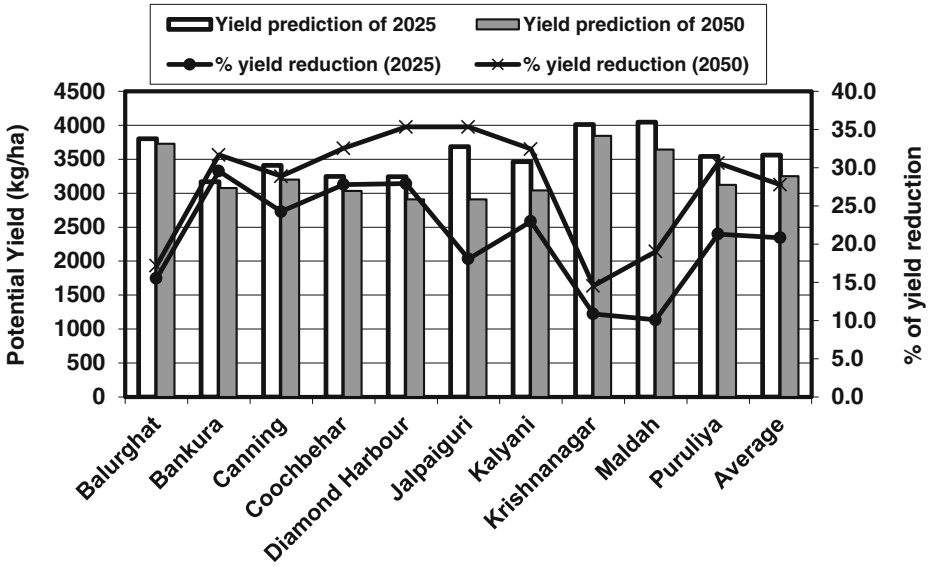


Fig. 1 The yield prediction of wet season rice in different stations in West Bengal, Eastern India

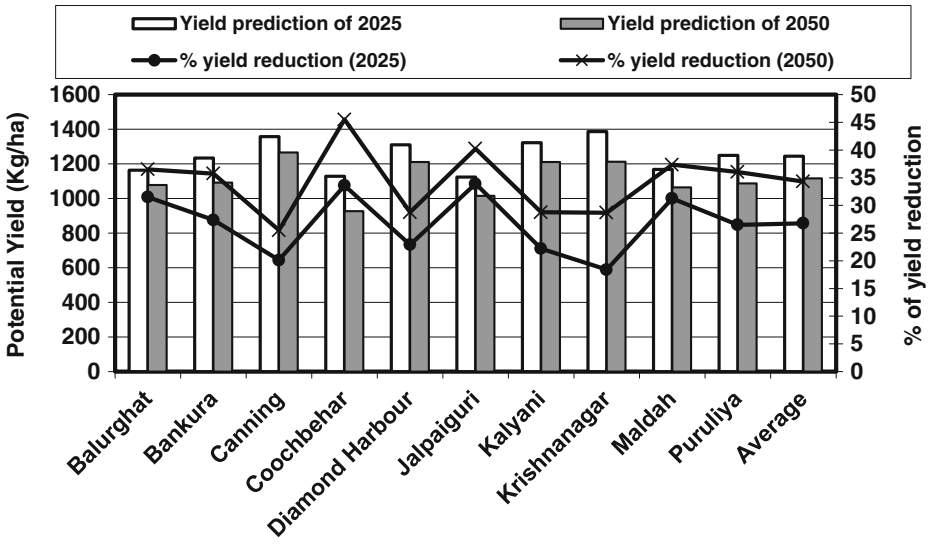


Fig. 2 The yield prediction of mustard in different stations in West Bengal, Eastern India

3.4 Adaptation strategies in view of climate change

Simple adaptation strategies involving adjustment of sowing time and seedling age at the time of transplanting were tested to improve the rice yield under climate change condition. It is observed that if older seedlings (about 25 days old) are transplanted under the condition of 1 °C temperature increase, the yield is further reduced. The optimum sowing time should be around 1st June and the late sowing beyond 15th June may be avoided in future (Table 5). With the help of CGSM, the effect of different nutrient dose on rice yield under elevated thermal condition was also evaluated. The normal nutrient dose (nitrogen: phosphorus: potassium or N: P: K) for wet-season rice is 80: 40: 40 kg ha $^{-1}$ in major parts of West Bengal. It was observed that the higher N: P: K dose (120: 40: 40 kg ha $^{-1}$) can solve the problem of temperature enhancement to some extent. Under elevated temperature condition (3 °C more), the normal nutrient dose resulted about 3,455.2 kg ha $^{-1}$ yield, whereas application of 120 kg ha $^{-1}$ nitrogen resulted 4,304.6 kg ha $^{-1}$ yield i.e. 24.6 % more yield (Fig. 3). Shrestha et al. (2014) also opined that higher dose of nutrient application can enhance the rice yield from 0.3 to 29.8 % in Central Vietnam under future climate.

In Eastern India, mustard is generally sown during mid October. But simulation under elevated thermal condition showed that the November-sown mustard performed better (Table 6). For the same crop, simulation was carried out with the increased seed rate (8 kg ha $^{-1}$ instead of 6 kg ha $^{-1}$),

Table 5 Effect of date of sowing and seedling-age on yield of wet-season rice under elevated temperature (1 °C rise over normal temperature) condition

Seedling-age (days) at the time of transplanting	Yield (kg ha $^{-1}$) for different transplanting date				
	23rd May	1st June	8th June	15th June	30th June
18	3,825.0	3,997.6	3,624.1	3,421.8	3,131.5
25	3,659.1	3,825.2	3,506.7	3,594.7	3,085.6

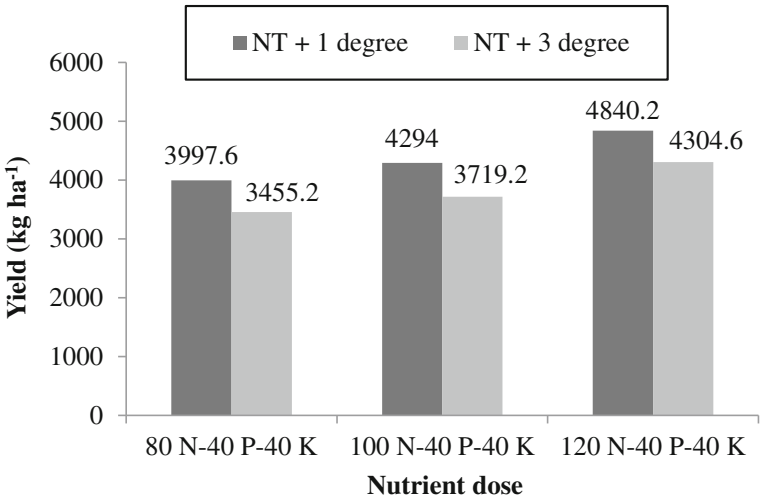


Fig. 3 The yield of wet-season rice for different nutrient doses under elevated thermal regime (NT = Normal temperature)

but the yield increase was negligible except for October-sown crop (Table 6). During October, the residual soil moisture, remaining after harvest of wet-season rice, can be utilized by mustard, hence the higher seed rate resulted higher yield for October-sown mustard.

4 Discussion

The high R^2 value, low RMSE and low SE indicate that the InfoCrop model can be used safely for yield prediction and impact assessment. The model can also simulate the biomass and crop phonology for both the crops under study (not presented in the paper).

For wet-season rice, the effect of 1 °C rise in normal temperature may not be so pronounced but for 3 °C rise the yield reduction would be as high as 21.1 %. If A2 scenario persists in Eastern India, the region may suffer from availability of staple-food in the year 2050 unless proper adaptation strategies are not taken. There will be a reduction in leaf area index and the grain numbers with increasing temperatures, which may be main cause of yield reduction. Sterility of rice spikelets at higher temperatures may be another reason. The yield reduction of rice in future would be higher in the drier tract of Red Lateritic zone of West Bengal. The said zone has the inherent problem of poor water holding capacity; hence the temperature rise will lead to detrimental effect on water availability to the crop. The simplest adaptation measures

Table 6 Effect of date of sowing on yield of mustard under elevated temperature (1 °C rise over normal temperature) condition

Seed rate	Yield (kg ha $^{-1}$) for different transplanting date				
	22nd Oct	29th Oct	4th Nov	11th Nov	18th Nov
6 kg ha $^{-1}$	1,277.7	1,221.8	1,345.3	1,298.7	1,205.4
8 kg ha $^{-1}$	1,348.6	1,225.2	1,198.6	1,079.5	1,027.3

should be the adjustment of sowing time, which can be scheduled during June first week for wet-season rice. The older seedling at the time of transplanting of rice may not be fruitful climate change adaptive measures as the crop growth duration is further shortened if seedlings of about 25 days older are transplanted. The application of 120: 40: 40 kg ha $^{-1}$ N: P: K proves to yield better than lower nutrient dose (80: 40: 40 kg ha $^{-1}$ N: P: K). The nitrogenous nutrient promotes the vegetative growth and increase the vigor of the plant. The higher leaf area promotes 'biomass to grain conversion' efficiency, resulting higher yield. Proper combination of organic and inorganic nutrients should be applied to sustain the better soil health and to reduce leaching and volatilization losses.

The yield reduction of mustard (25.0 %) due to 1 °C rise in temperature is much higher than that of wet-season rice. It may be as high as 32.7 % for 3 °C rise in temperature. The rice is grown under puddled condition where stagnant water is maintained throughout the growing period. This may reduce the negative impact of temperature enhancement. As under 1 °C temperature rise, the crop growth period of rice was reduced by only 2 days, the effect of reduced crop duration on yield is not so intense (8.7 % only). In case of mustard, the temperature enhancement will reduce the crop growth period up to 5 days for 1 °C rise which may be one of the causes for 25.5 % yield reduction. The temperature enhancement during winter season may also cause water shortage leading to poor yield of mustard. As per PRECIS output, Terai zone of West Bengal will receive lower amount of winter rainfall, which may be the reason for higher magnitude of yield reduction of mustard in this zone. Like rice, sowing time adjustment is the simplest adaptation measures also for mustard. The optimum sowing time for the study area should be November first week. The higher seed rate (8 kg ha $^{-1}$) of mustard can be used only if October sowing is done. For November-sown mustard, 6 kg ha $^{-1}$ seed rate will be the effective adaptation measure.

The Working Group II contribution to the IPCC's Fifth Assessment Report (WG II AR5) recognizes that the risk of climate change will vary across regions and adaptation will vary as per social and economical drivers of vulnerability (IPCC 2014). The report includes wide array of adaptation options, namely, changes in land use, such as agroforestry (Verchot et al. 2007); conservation agriculture and increase soil organic matter (Clough et al. 2011); new crop varieties and genetic techniques; efficient irrigation and water saving technologies (Game et al. 2011). The present paper adds up simple agronomic adjustment as adaptation options, which will be very much useful for sustaining the food production in Eastern India under elevated thermal situation.

5 Conclusions

It can be concluded that the probable change in climate will have a negative impact on crop growth and yield in Eastern India. The negative impact of climate change on mustard grown in winter season will be more pronounced compared to wet-season rice. The projected yield reduction for 2025 and 2050 over West Bengal state is not uniform due to wide variation of local features and future climatic scenario. To cope up with the adverse thermal situation in future, the following adaptation options should be followed in Eastern India:

i) For wet-season rice: It should be transplanted during June first week. For late transplanting, yield reduction will be more under elevated thermal condition. The seedling age should be 18 days at the time of transplanting of wet-season rice. The nutrient dose of 120: 40: 40 kg ha $^{-1}$ N: P: K will be more effective to combat the ill effect of high temperature.

ii) For mustard: It should be sown during November first week. If it is sown in October, increased seed rate (8 kg ha $^{-1}$ instead of 6 kg ha $^{-1}$) should be used.

This paper leaves the scope of evaluating such types of simple agronomic adjustment as climate change adaptation options for other crops.

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